

Blinking LED Project

Picobricks

Beginner

10 Mins

1 Overview

In computer programming, when learning a new coding language we typically create a **Hello World** project, which prints **Hello World** to the screen.

When working with robotics and physical computing, our **Hello World** project becomes a blinking **LED** (light-emitting diode).

Let's take a look at how we'd create this on the PicoBricks using the Brick IDE.



What you'll learn

- How to make an LED blink on and off

2 What you'll need



What you'll need

- PicoBricks Kit
- PicoBricks Brick IDE
- MicroUSB to USB cable
- Laptop or tablet for coding

3 Key words

Key words

LED

An LED is a tiny light you can switch on and off (and sometimes change colour) with your code. LED is short for "light-emitting diode".

The little standby light on a TV is an LED, your code can flash one on and off the same way.

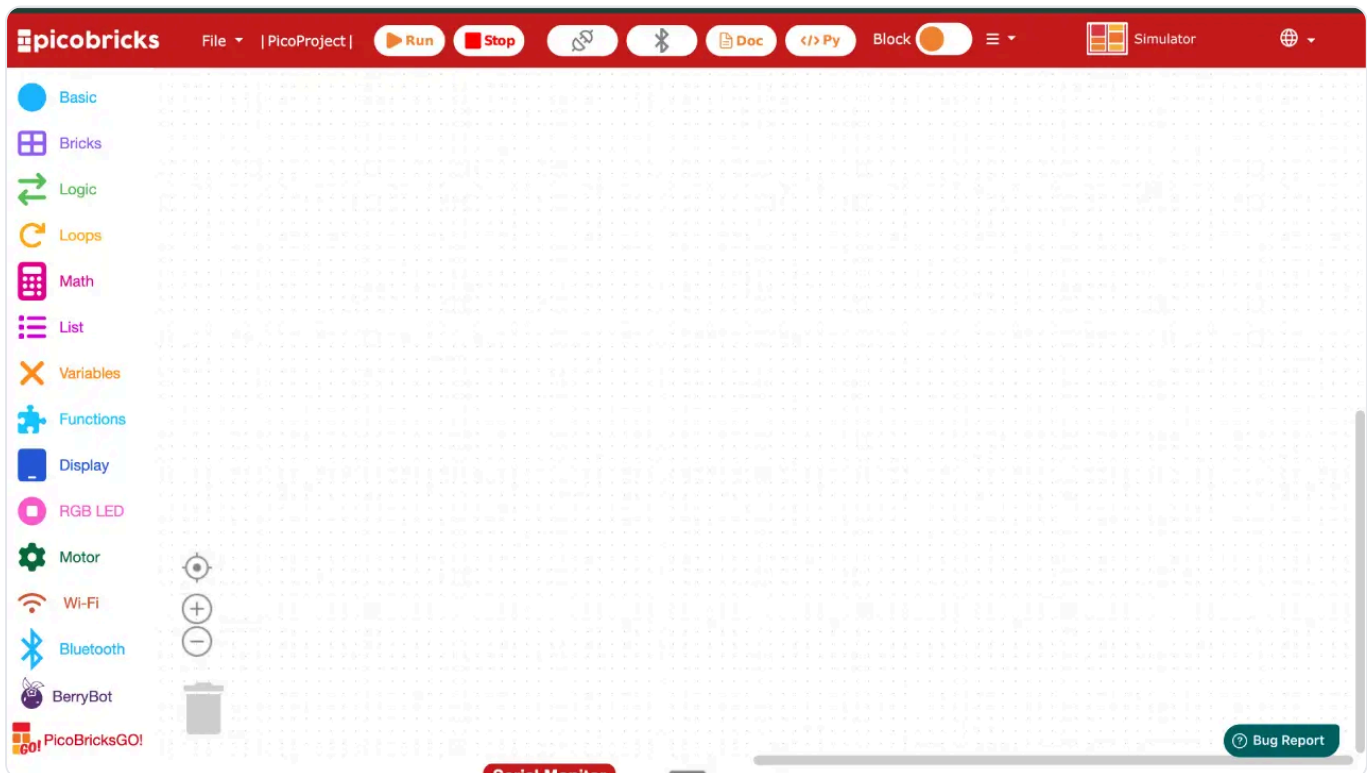
Forever loop

A forever loop repeats its instructions again and again without ever stopping (until you switch off or reset). It's perfect for anything you want to keep happening over and over.

Like a clock's second hand that keeps ticking round and round, a forever loop keeps running the same steps over and over.

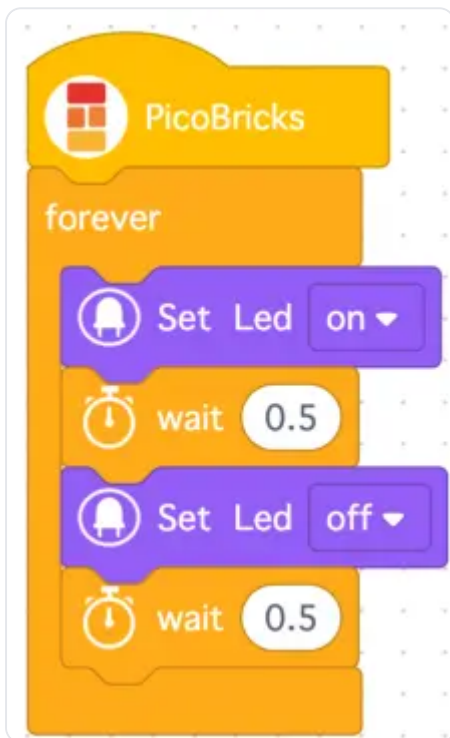
4 Set up your PicoBricks

1. Open your favourite web browser. We recommend either Google Chrome or Microsoft Edge. Type ide.picobricks.com into the address bar.
2. Connect the PicoBricks board to your computer using the USB cable.
3. Connect the PicoBricks board to the Brick IDE by selecting the **Connection** button at the top of the code area.



5 Build the code

1. Select **Basic**. Select a `PicoBricks` block and drop it in the code area.
2. Select **Loops**. Select a `forever` block and snap it to the `PicoBricks` block.
3. Select **Bricks**. Select a `Set Led` block and snap it within the `forever` block. It will be set to **on** by default.
4. Select **Loops**. Select a `wait` block and snap it to the `Set Led on` block. Change the **1** to **0.5**.
5. Select **Bricks**. Select a `Set Led` block and snap it to the `wait 0.5` block. Change the **on** to **off**.
6. Select **Loops**. Select a `wait` block and snap it to the `Set Led off` block. Change the **1** to **0.5**.

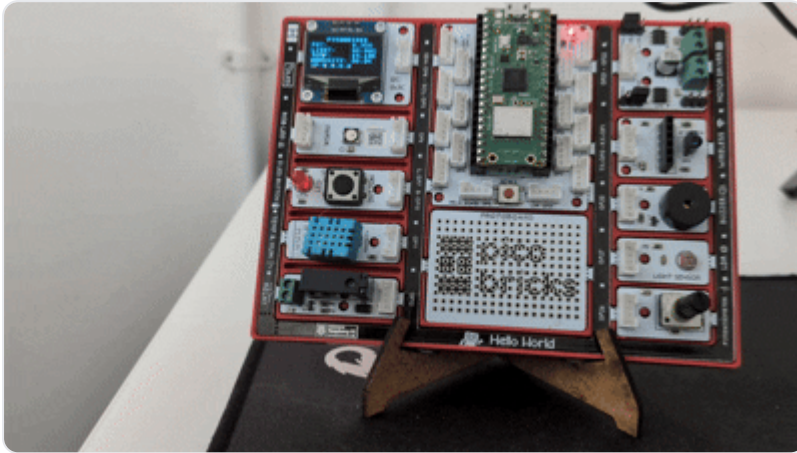


We've now completed the code for our blinking LED.

6 Run it and watch

1. Select the **run** button at the top of the code window.

Your LED should now blink on and off every half a second. 🎉



7 Try it yourself

🚀 Try it yourself

Challenge: Can you make your LED blink faster or slower? Try changing both `wait 0.5` blocks to a smaller number like `0.1` for a speedy flash, or a bigger number like `1` for a slow, sleepy blink. What happens if you give the two waits different numbers?

8 Stuck? Quick fixes

Problem	Try this
The LED isn't lighting up at all	Check the USB cable is pushed in properly at both ends, and make sure the Brick IDE shows it's connected to your PicoBricks board.
The blocks won't snap together	Make sure each block is dropped just inside the one above it. The <code>Set Led</code> and <code>wait</code> blocks need to sit inside the <code>forever</code> loop, not below it.
The LED comes on but never turns off	Check your second <code>Set Led</code> block is set to off , and that a <code>wait</code> block sits after each <code>Set Led</code> block.
Nothing happens when I press run	Try reconnecting the board, then press run again. Unplugging the USB cable and plugging it back in often helps too.